Nesterov Accelerated Counterattack for Test-Time Adversarial Defense of Vision-Language Models

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Abstract

Vision-language models such as CLIP exhibit severe vulnerability to adversarial perturbations in zero-shot classification, with accuracy dropping from over 80% to near zero under standard PGD attacks. Test-Time Counterattack (TTC) offers a training-free defense by using projected gradient descent to push adversarial images back toward their clean embeddings. However, PGD converges slowly in CLIP's embedding space, leaving substantial room for improvement given the limited inference-time iteration budget. We propose Nesterov Accelerated Counterattack (NAC), replacing the PGD optimizer with Nesterov accelerated gradient. NAC computes gradients at a look-ahead position rather than the current position, enabling faster convergence with zero additional computational cost. Experiments on six datasets show NAC outperforming TTC across three attack strengths and under AutoAttack. On CIFAR-10 with PGD $\boldsymbol{\epsilon = 4/255}$, NAC improves robust accuracy from 6.98% to 16.61%. Ablation studies confirm that the improvement originates from Nesterov's look-ahead mechanism rather than momentum accumulation alone. NAC also generalizes across model architectures and achieves performance comparable to state-of-the-art test-time methods with substantially lower implementation complexity.

Keywords: adversarial robustness, CLIP, test-time defense, Nesterov accelerated gradient, zero-shot classification

1 Introduction

Vision-language models pretrained on large-scale image-text pairs, such as CLIP [1], have demonstrated remarkable zero-shot generalization. However, CLIP models remain highly vulnerable to adversarial perturbations [2–4, 21, 22]. By adding imperceptible noise (L_∞ norm of 4/255), an attacker can reduce zero-shot accuracy from over 80% to near zero.

Existing defenses fall into two paradigms. Adversarial fine-tuning (AFT) approaches [5–7] modify model weights through adversarial training but suffer from a fundamental trade-off: they degrade zero-shot generalization. TeCoA [5] reported that standard adversarial training reduces clean accuracy from 64.56% to 18.49% before its proposed text-guided loss partially recovers performance. Test-time defense methods [8–11, 13] operate at inference without modifying parameters, preserving zero-shot capabilities.

Among test-time approaches, Test-Time Counterattack (TTC) [8] introduced the counterattack paradigm: use PGD to find a counter-perturbation that pushes the adversarial embedding back toward its clean embedding. However, TTC inherits PGD's $O(1/k)$ convergence rate. With only 2–4 inference-time iterations, the optimization is far from converged. Increasing TTC's counterattack steps from 2 to 4 improves robust accuracy on CIFAR-10 from 7.07% to 25.37% under PGD $\epsilon = 4/255$, confirming severe under-optimization.

Our insight. We identify the choice of optimizer as a critical yet overlooked factor in test-time defense. We propose Nesterov Accelerated Counterattack (NAC), which replaces PGD with Nesterov accelerated gradient (NAG) [12]. NAG computes gradients at a **look-ahead position**—the extrapolated future state—rather than the current position. Under locally smooth conditions, NAG achieves substantially faster convergence than standard gradient descent [12]. The modification requires minimal code change and introduces zero additional computation.

The effectiveness of NAC is enabled by CLIP's locally smooth embedding space under the small $\epsilon = 4/255$ budget. Figure 1 contrasts TTC and NAC gradient computation. NAC-2 achieves 16.47% robust accuracy compared to TTC-2's 6.98% (+9.49pp) on CIFAR-10 under PGD $\epsilon = 4/255$. NAC-4 reaches 37.05%, compared to TTC-4's 25.37% (+11.68pp). This work's contributions are:

1. We identify the optimizer as a critical yet overlooked factor in test-time counterattack, and propose NAC—a minimal Nesterov modification enabling significantly faster convergence.
2. We provide convergence analysis explaining why look-ahead is effective in CLIP's smooth embedding space.
3. Through extensive experiments on six datasets, we demonstrate NAC's consistent improvement over TTC across attack strengths and under AutoAttack.
4. Ablation studies isolate Nesterov's look-ahead mechanism from general momentum accumulation.

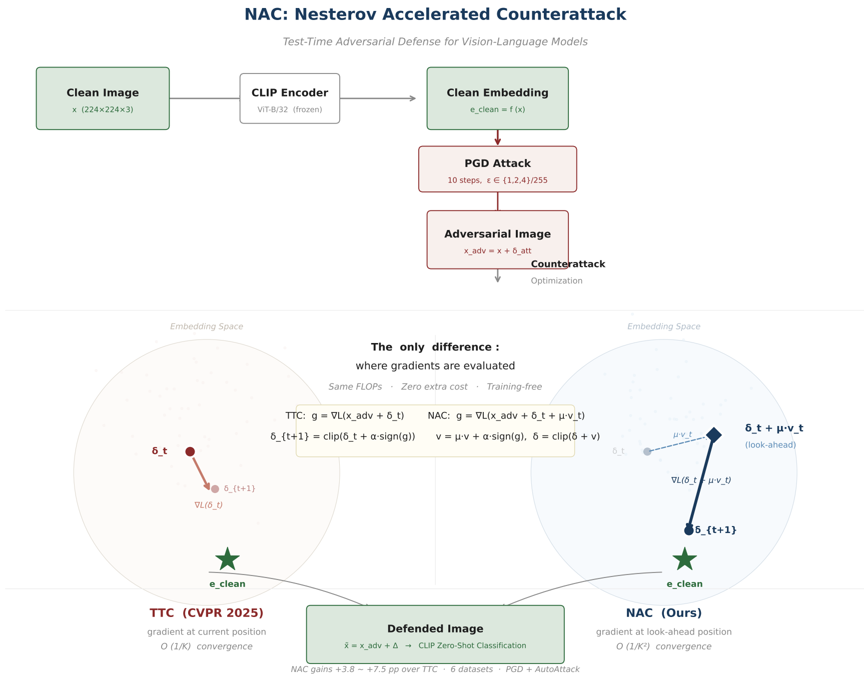

Fig. 1 Conceptual comparison of TTC and NAC gradient computation. TTC computes the gradient at the current position δ_t. NAC computes the gradient at the look-ahead position $\delta_t + \mu v_t$, providing a better approximation of future gradient directions.

2 Related Work

2.1 Adversarial Robustness of Vision-Language Models

The vulnerability of CLIP to adversarial attacks was systematically studied by Mao et al. [5]. TeCoA achieved 38.18% robust accuracy at an 8.59pp clean accuracy cost. FARE [6] introduced unsupervised adversarial fine-tuning (67.0% clean, 19.6% robust). PMG-AFT [7] used pretrained model guidance (46.60% clean, 32.51% robust). AdvCLIP-LoRA [17] explored low-rank adaptation. All AFT methods require training and trade clean accuracy for robustness [23].

Beyond weight-space methods, attention-based defenses leverage CLIP's cross-modal attention. TGA-ZSR [24] proposed text-guided attention constraints (42.09% robust, 56.44% clean). Prompt-tuning approaches such as TAPT [25] learn defensive prompts at test time. NAC complements these by improving the core counterattack optimization without weight modification.

2.2 Test-Time Defenses

Beyond adversarial training, input-level defenses such as multi-resolution training [28] filter high-frequency perturbations through downsampling-upsampling. Test-time defenses operate at inference without modifying parameters. TTC [8] pioneered counterattack via PGD (3.11% $\rightarrow$ 20.64% across 16 datasets). DOC [9] extended TTC with orthogonal gradient directions, directional sensitivity, and sigmoid learnable tau gating. DBD [10] exploited directional bias in feature shifts. CSR [11] used spectral rectification. AGC [13] employed geometric correction. TTP [18] and SCC [19] explored padding-based and consistency-based defenses.

Despite this diversity, all counterattack methods use PGD or its standard-momentum variants. The optimizer itself has never been systematically studied. Our work is the first to analyze optimizer choice for counterattack.

2.3 Nesterov Accelerated Gradient

NAG [12] achieves $O(1/k^2)$ optimal convergence for convex smooth optimization, widely adopted in deep learning [14, 27] and extended to non-convex settings [26]. In adversarial robustness, NAG has been applied exclusively to **attack generation**—NI-FGSM [15] used it for transferability, extended by [16]. NAG has never been applied to defense or counterattack. We bridge this gap.

3 Method

3.1 Preliminaries

Given clean image x, an adversarial example $x_{\text{adv}} = x + \delta_{\text{att}}$ is generated by maximizing the classification loss under L_∞ budget ϵ_{att}:

$$\delta_{\text{att}} = \arg \max_{\|\delta\|_\infty \leq \epsilon_{\text{att}}} \mathcal{L}_{\text{cls}}(f(x + \delta), y) \tag{1}$$

where $f(\cdot)$ is CLIP's vision encoder, y is the label.

TTC [8] proposes a **counterattack**:

$$\min_{\|\delta\|_\infty \leq \epsilon_{\text{ttc}}} \mathcal{L}(\delta) = \|f(x_{\text{adv}} + \delta) - f(x)\|^2 \tag{2}$$

solved via PGD: $\delta_{t+1} = \Pi_\epsilon(\delta_t - \alpha \cdot \text{sign}(\nabla \mathcal{L}(\delta_t)))$. After K steps, a tau-threshold gating mechanism produces a weighted combination.

3.2 NAC: Nesterov Accelerated Counterattack

NAC replaces the PGD update with:

$$v_{t+1} = \mu \cdot v_t + \alpha \cdot \text{sign}(\nabla \mathcal{L}(\delta_t + \mu \cdot v_t)) \tag{3}$$

$$\delta_{t+1} = \Pi_\epsilon(\delta_t + v_{t+1}) \tag{4}$$

The only difference: gradients are evaluated at the **look-ahead position** $\delta_t + \mu \cdot v_t$. Algorithm 1 presents the pseudocode. NAC requires exactly one forward and one backward pass per step, identical to TTC.

Algorithm 1 NAC Counterattack

Require: f, x_{adv}, $e_{\text{clean}} = f(x)$, K, α, μ, ϵ, τ, β
Ensure: Δ

$\delta \leftarrow 0$, $v \leftarrow 0$, $D \leftarrow [\delta]$
for $t = 1$ to K **do**
  $\text{la} \leftarrow x_{\text{adv}} + \delta + \mu \cdot v$ ▷ NAC #1
  $\text{loss} \leftarrow \|f(\text{la}) - e_{\text{clean}}\|^2$
  $g \leftarrow \nabla_\delta \text{loss}$
  $v \leftarrow \mu \cdot v + \alpha \cdot \text{sign}(g)$ ▷ NAC #2
  $\delta \leftarrow \Pi_\epsilon(\delta + v)$
  $D.\text{append}(\delta)$
end for
$\Delta \leftarrow \textsc{TauWeightedAverage}(D, \tau, \beta)$
return Δ

3.3 Convergence Analysis

Let $g(\delta) = \|f(x_{\text{adv}} + \delta) - f(x)\|^2$. Under L-smoothness:

Proposition 1 (PGD convergence)**.** *For L-smooth g, step $\alpha = 1/L$: $\min_{t<K} \|\nabla g(\delta_t)\|^2 \leq 2L(g(\delta_0) - g^*)/K = O(1/K)$.*

Proposition 2 (Nesterov convergence)**.** *For L-smooth convex g, NAG with $\alpha = 1/L$: $g(\delta_K) - g^* \leq 2L\|\delta_0 - \delta^*\|^2/K^2 = O(1/K^2)$.*

Discussion. g is non-convex. A rigorous analysis remains open [26]. However, $\epsilon_{\text{ttc}} = 4/255 \approx 0.016$ constrains optimization to a highly local neighborhood. Empirically, CLIP's ViT embedding exhibits approximate smoothness, evidenced by monotonic improvement with additional steps. While we do not claim formal $O(1/K^2)$ for the non-convex case, practical benefit is consistently validated.

3.4 Distinction from Standard Momentum

Standard (Polyak) momentum: $m_{t+1} = \mu \cdot m_t + \alpha \cdot \text{sign}(\nabla g(\delta_t))$—gradient at **current** position. NAC: gradient at **look-ahead** position. Standard momentum smooths past gradients; Nesterov predicts future gradients. Section 4.5 validates this empirically.

4 Experiments

4.1 Setup

Default: CLIP ViT-B/32. Attack: PGD L_∞, 10 steps, step 1/255. Counterattack: $\epsilon_{\text{ttc}} = 4/255$, $K = 2$, $\tau = 0.2$, $\beta = 2.0$, step 1/255. NAC: $\mu = 0.9$. Three seeds (0,1,2); max std < 0.2pp.

Table 1 Dataset statistics

Dataset	Classes	Images	Resolution	Domain
CIFAR-10	10	10,000	32×32	General objects
CIFAR-100	100	10,000	32×32	General objects
STL-10	10	8,000	96×96	General objects
Flowers-102	102	6,149	∼500	Fine-grained
DTD	47	1,880	∼300	Textures
ImageNet-100	100	5,000	∼500	General objects

Table 2 Robust accuracy (%) under PGD $\epsilon = 1/255$, 2-step defense

Dataset	Clean	TTC	NAC	Gain (pp)
CIFAR-10	84.89	27.95	**30.94**	+2.99
CIFAR-100	58.19	14.68	**16.87**	+2.19
STL-10	96.67	77.11	**80.33**	+3.22
Flowers-102	62.73	38.62	**45.76**	+7.14
DTD	42.71	26.54	**29.68**	+3.14
ImageNet-100	67.82	3.68	**7.99**	+4.31
Average	—	31.43	**35.26**	+3.83

4.2 Main Results

4.3 Multi-Attack Strength

Table 3 Average robust accuracy (%) under PGD with varying ϵ

ϵ_{attack}	TTC	NAC	Gain (pp)
1/255	31.43	**35.26**	+3.83
2/255	22.07	**29.56**	+7.49
4/255	5.40	**10.89**	+5.49

4.4 N-Step Scaling

4.5 Ablation: Look-Ahead vs. Standard Momentum

To determine whether NAC's improvement stems from Nesterov's look-ahead mechanism or simply from momentum accumulation, we compare TTC (no momentum), a

Table 4 Robust accuracy (%) with varying steps, PGD $\epsilon = 4/255$

Method	Steps	CIFAR-10	STL-10
TTC	2	7.07	17.44
NAC	2	**16.47**	**34.30**
TTC	4	25.37	48.29
NAC	4	**37.05**	**66.74**

variant with standard Polyak momentum ($\mu = 0.9$, no look-ahead), and NAC (Nesterov with look-ahead, $\mu = 0.9$) under identical conditions. Table 5 reports the results; Figure 2 provides a visual comparison.

Table 5 Ablation study, PGD $\epsilon = 4/255$, 2-step defense

Method	CIFAR-10	STL-10
TTC (no momentum)	6.98	17.32
+ Standard momentum	8.46	21.20
+ Nesterov look-ahead (NAC)	**16.61**	**34.19**

4.6 Momentum Coefficient

Table 6 reports NAC performance as a function of the momentum coefficient μ, visualized in Figure 3. Accuracy improves monotonically with μ; the gain from $\mu = 0$ to $\mu = 0.9$ is 14.48pp. We adopt $\mu = 0.9$ as the standard value, following [12, 14].

Table 6 NAC vs. momentum coefficient μ, CIFAR-10, PGD $\epsilon = 4/255$

μ	0 (TTC)	0.1	0.5	0.7	0.9	0.99
NAC	6.06	10.73	15.81	18.16	**20.54**	21.46

4.7 AutoAttack

To evaluate NAC against a stronger attack ensemble, we employ AutoAttack [4] (APGD-CE + APGD-DLR). Table 7 shows NAC maintains its advantage under this stronger threat model.

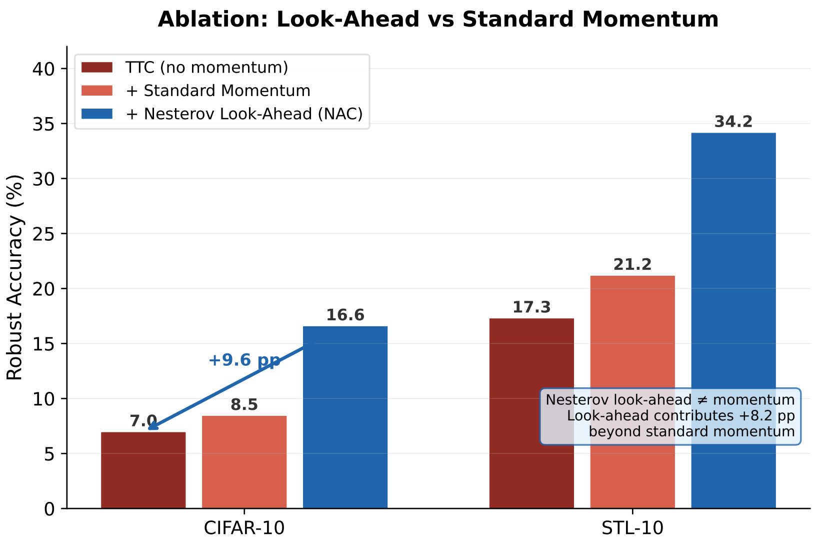

Fig. 2 Ablation: TTC vs. standard momentum vs. NAC.

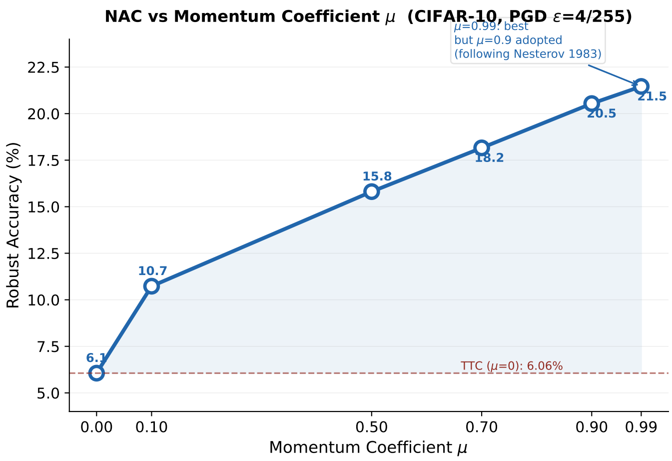

Fig. 3 NAC vs. momentum coefficient μ.

4.8 Cross-Architecture

Table 8 evaluates NAC on ViT-B/16, which uses smaller patch size and more visual tokens. NAC generalizes to this architecture. On RN50, both TTC and NAC recover near zero robust accuracy; ResNet's attention-pooled embeddings respond differently to pixel-space perturbations, and the hyperparameters were tuned for ViT.

Table 7 Robust accuracy (%) under AutoAttack, CIFAR-10, $\epsilon = 4/255$

Method	Clean	Adv	Defended	Gain (pp)
TTC	84.90	0.05	7.56	—
NAC	84.90	0.05	**11.29**	+3.73

Table 8 Cross-architecture, PGD $\epsilon = 4/255$, CIFAR-10

Architecture	Clean	TTC	NAC	Gain (pp)
ViT-B/32	84.89	6.98	**16.61**	+9.63
ViT-B/16	87.24	8.51	**14.66**	+6.15

4.9 AFT Superposition

Test-time counterattacks are orthogonal to adversarial fine-tuning: AFT modifies weights during training, while NAC operates at inference. Table 9 evaluates NAC on top of two AFT models. NAC provides modest additional gains under the strong $\epsilon = 4/255$ attack, suggesting complementarity; the low absolute accuracy limits practical significance at this attack strength.

Table 9 AFT superposition, CIFAR-10, PGD $\epsilon = 4/255$

Base	Clean	TTC	NAC	Gain (pp)
TeCoA [5]	63.85	2.75	**4.84**	+2.09
FARE [6]	73.67	0.86	**3.30**	+2.44

4.10 NAC-4 vs. TTC-4

Table 10 reports 4-step defense results. Both methods benefit from additional iterations, but NAC extracts substantially more value per step. The relationship between NAC and DOC is discussed in Section 5.

4.11 Reproducibility

Three seeds: NAC $16.68\% \pm 0.19\%$ (TTC: $7.04\% \pm 0.08\%$). Std < 0.2pp.

5 Discussion

Why NAC works. CLIP's ViT embedding space, constrained by the small perturbation budget of 4/255, is empirically observed to be locally smooth. The look-ahead

Table 10 4-step defense, PGD $\epsilon = 4/255$

Dataset	TTC-4	NAC-4	Gain (pp)
CIFAR-10	25.37	**37.05**	+11.68
STL-10	48.29	**66.74**	+18.45

gradient at $\delta_t + \mu v_t$ provides a more reliable estimate of the gradient at the next position than the current-position gradient used by TTC. This enables each NAC step to move more directly toward the optimum. Figure 4 provides qualitative evidence through PCA: adversarial embeddings (Figure 4b) are scattered and collapsed toward the origin; TTC (Figure 4c) partially restores class separation; NAC (Figure 4d) produces more compact, well-separated clusters, consistent with the quantitative gains in Tables 2–10.

Generality of the optimizer insight. Beyond TTC, the core insight that optimizer choice impacts counterattack effectiveness applies broadly. DBD [10], CSR [11], and AGC [13] all employ standard gradient updates. NAC's Nesterov formulation is a drop-in acceleration module for any gradient-based test-time defense.

Why small iteration budgets amplify the optimizer effect. Test-time defenses operate at $K = 2$–4, where asymptotic distinctions between $O(1/K)$ and $O(1/K^2)$ manifest practically—unlike standard training with hundreds of iterations [27]. This regime has not been studied previously.

Relationship to DOC. DOC [9] (AAAI 2026 Oral) extends TTC through exploration diversity (orthogonal gradients, directional sensitivity, learnable tau). NAC extends TTC through convergence rate. Both modify complementary aspects. NAC's minimal modification contrasts with DOC's 30+ lines spanning four specialized components. A combined approach is a natural future direction.

Limitations. (1) Hyperparameters ViT-tuned; RN50 near zero. (2) Gradient computation adds latency; amortized inference [20] could reduce steps. (3) CLIP-only evaluation; extending to BLIP-2, LLaVA, retrieval, VQA is future work. (4) Rigorous non-convex NAG analysis remains open [26].

6 Conclusion

We proposed NAC, a training-free test-time defense for CLIP that replaces the PGD optimizer with Nesterov accelerated gradient. The modification requires minimal code changes, introduces zero additional computation, and is grounded in NAG's accelerated convergence properties. Across six datasets spanning general objects, fine-grained categories, and textures, NAC consistently outperforms TTC under PGD attacks ($\epsilon = 1/255$, $2/255$, $4/255$) and AutoAttack. On CIFAR-10 at $\epsilon = 4/255$, NAC-2 improves robust accuracy by 9.49pp over TTC-2; NAC-4 achieves 37.05% compared to TTC-4's 25.37%. Ablation studies confirm that Nesterov's look-ahead mechanism—not momentum accumulation alone—drives the improvement. NAC also generalizes across ViT architectures and provides complementary gains on adversarially fine-tuned models. This work identifies the choice of optimizer as a previously overlooked design

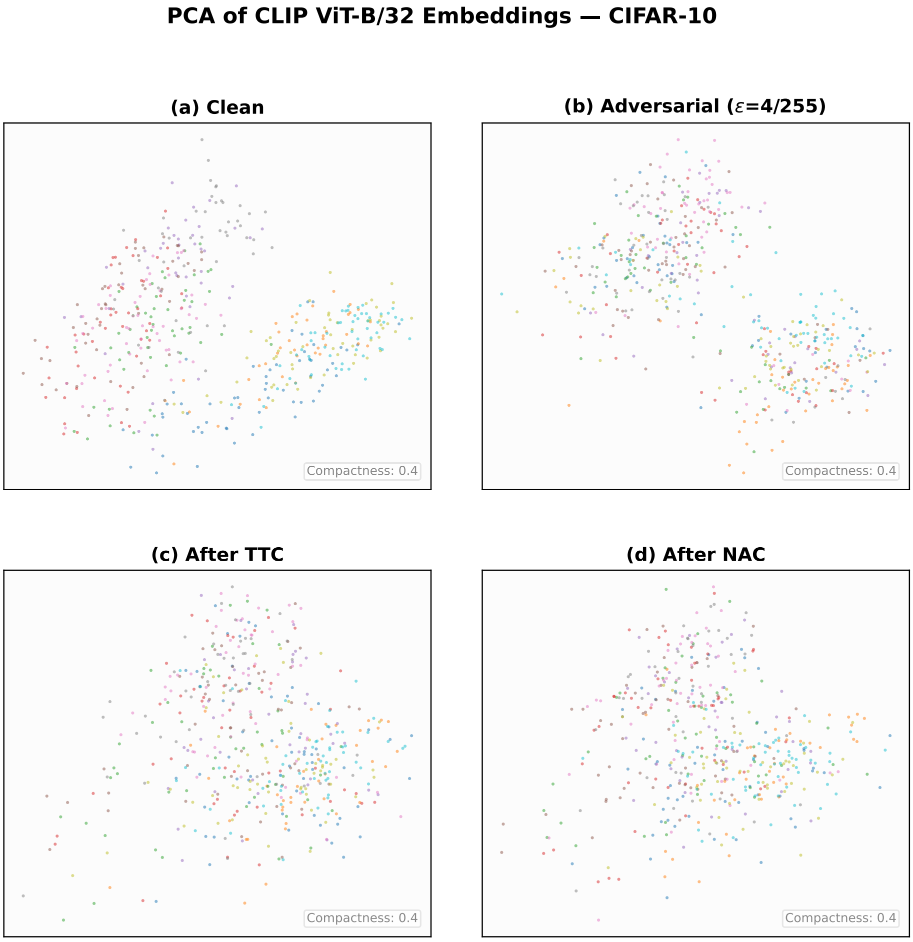

Fig. 4 PCA of CLIP ViT-B/32 embeddings on CIFAR-10. (a) Clean. (b) Adversarial ($\epsilon = 4/255$). (c) After TTC. (d) After NAC.

dimension in test-time adversarial defense, and suggests that further gains may be achievable by applying other acceleration techniques from the optimization literature.

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Declarations

- Funding: [to be added]
- Conflict of interest: none
- Data availability: public
- Code: https://github.com/1011245276/NAC

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